

Customer Churn Data Exploration

Source file: customer_churn_dataset-testing-master.csv

Rows: 64,374 data rows

Columns: 12

Duplicate rows: 0

Numeric columns: 9 (CustomerID, Age, Tenure, Usage Frequency, Support Calls, Payment Delay, Total Spend, Last Interaction, Churn)

Categorical columns: 3 (Gender, Subscription Type, Contract Length)

Missing values means true null/NA values detected by pandas. Blank values means empty or whitespace-only strings.

Column Data-Quality Overview

Column	Data type	Non-null	Nulls	Blank strings	Unique values	Missing %
CustomerID	int64	64374	0	0	64374	0.0
Age	int64	64374	0	0	48	0.0
Gender	str	64374	0	0	2	0.0
Tenure	int64	64374	0	0	60	0.0
Usage Frequency	int64	64374	0	0	30	0.0
Support Calls	int64	64374	0	0	11	0.0
Payment Delay	int64	64374	0	0	31	0.0
Subscription Type	str	64374	0	0	3	0.0
Contract Length	str	64374	0	0	3	0.0
Total Spend	int64	64374	0	0	901	0.0
Last Interaction	int64	64374	0	0	30	0.0
Churn	int64	64374	0	0	2	0.0

Numeric Descriptive Statistics

Column	count	mean	std	min	25%	50%	75%	max
CustomerID	64374.0	32187.5	18583.32	1.0	16094.25	32187.5	48280.75	64374.0
Age	64374.0	41.97	13.92	18.0	30.0	42.0	54.0	65.0
Tenure	64374.0	31.99	17.1	1.0	18.0	33.0	47.0	60.0
Usage Frequenc	64374.0	15.08	8.82	1.0	7.0	15.0	23.0	30.0
Support Calls	64374.0	5.4	3.11	0.0	3.0	6.0	8.0	10.0
Payment Delay	64374.0	17.13	8.85	0.0	10.0	19.0	25.0	30.0
Total Spend	64374.0	541.02	260.87	100.0	313.0	534.0	768.0	1000.0
Last Interaction	64374.0	15.5	8.64	1.0	8.0	15.0	23.0	30.0
Churn	64374.0	0.47	0.5	0.0	0.0	0.0	1.0	1.0

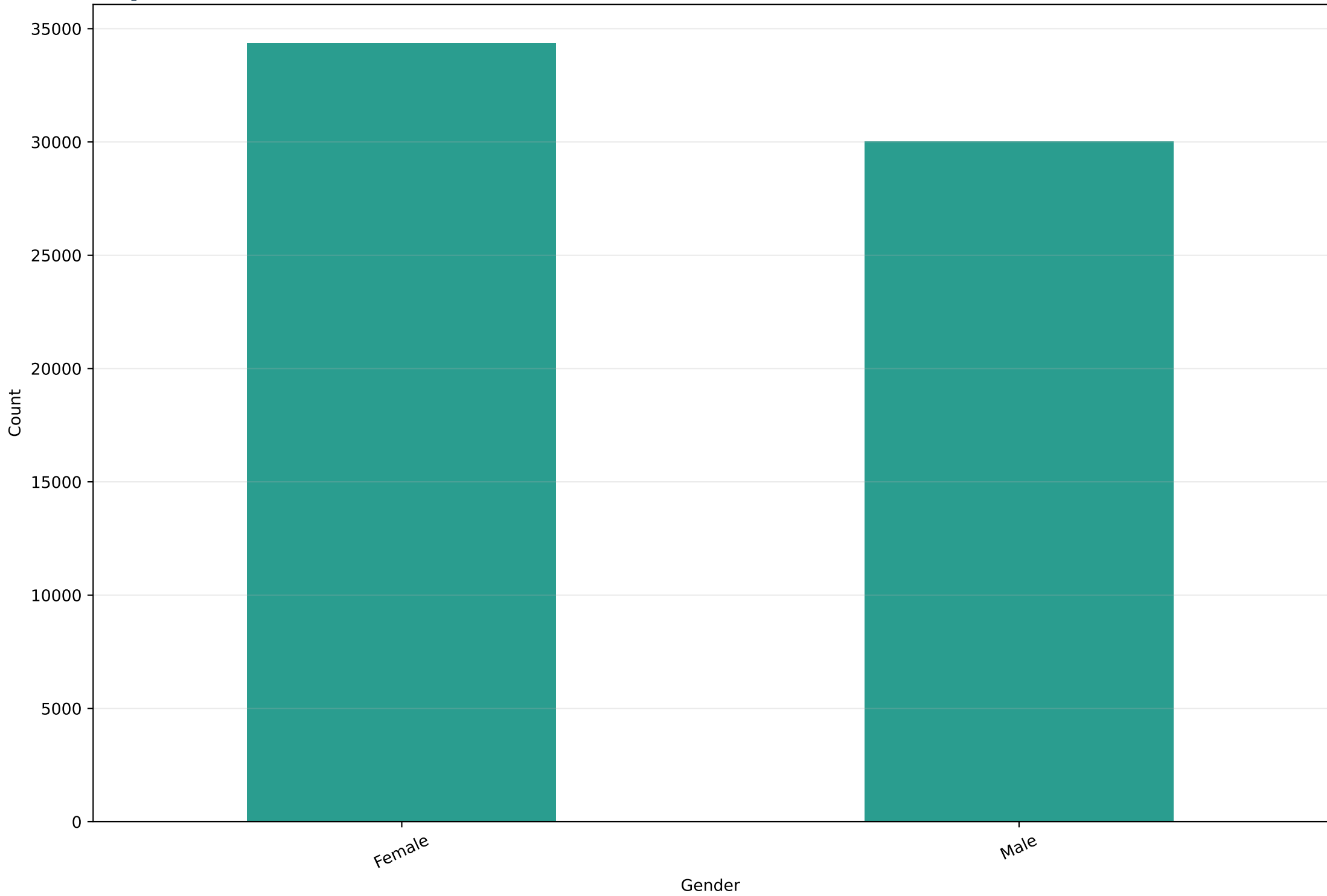
Numeric Ranges, Shape, and Special Values

Column	Minimum	Maximum	Median	Q1	Q3	Mean	Std dev	Zeros	Negative va	Skewness
CustomerID	1	64374	32187.5	16094.25	48280.75	32187.5	18583.32	0	0	-0.0
Age	18	65	42.0	30.0	54.0	41.97	13.92	0	0	-0.04
Tenure	1	60	33.0	18.0	47.0	31.99	17.1	0	0	-0.13
Usage Freq	1	30	15.0	7.0	23.0	15.08	8.82	0	0	0.04
Support Calls	0	10	6.0	3.0	8.0	5.4	3.11	4967	0	-0.19
Payment Del	0	30	19.0	10.0	25.0	17.13	8.85	1594	0	-0.35
Total Spend	100	1000	534.0	313.0	768.0	541.02	260.87	0	0	0.05
Last Interacti	1	30	15.0	8.0	23.0	15.5	8.64	0	0	0.01
Churn	0	1	0.0	0.0	1.0	0.47	0.5	33881	0	0.11

Numeric Averages by Churn Group

Feature	Churn = 0	Churn = 1
CustomerID	22846.79	42566.03
Age	41.13	42.9
Tenure	28.83	35.52
Usage Frequency	16.04	14.01
Support Calls	4.5	6.4
Payment Delay	12.45	22.33
Total Spend	560.54	519.34
Last Interaction	15.52	15.47
Churn	0.0	1.0

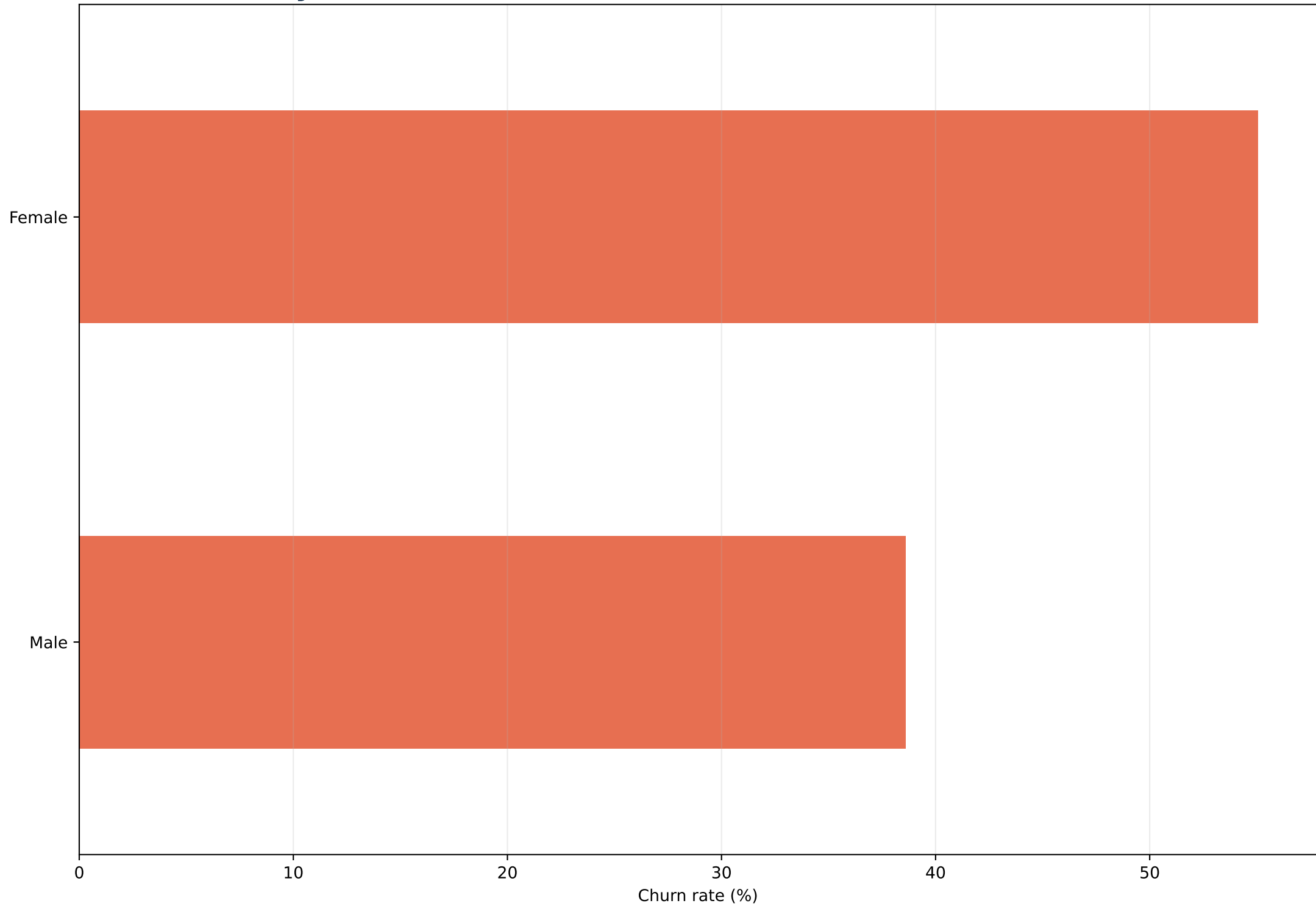
Top Values: Gender



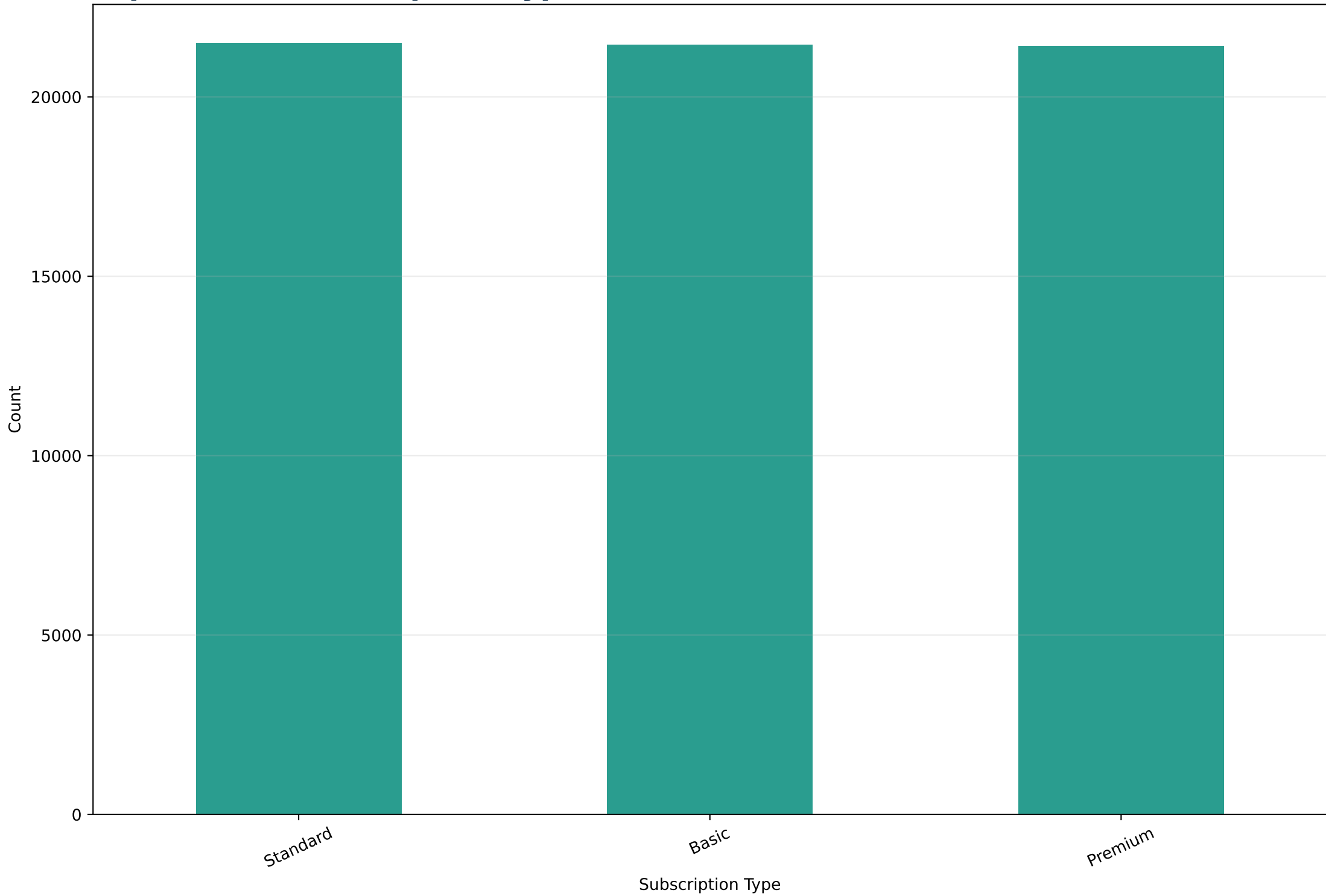
Churn Rate Detail: Gender

Gender	Customers	Churned	Churn rate %
Female	34353	18911	55.05
Male	30021	11582	38.58

Churn Rate by Gender



Top Values: Subscription Type



Churn Rate Detail: Subscription Type

Subscription Type	Customers	Churned	Churn rate %
Basic	21451	10356	48.28
Premium	21421	9960	46.5
Standard	21502	10177	47.33

Churn Rate by Subscription Type

Subscription Type

Basic

Standard

Premium

0

10

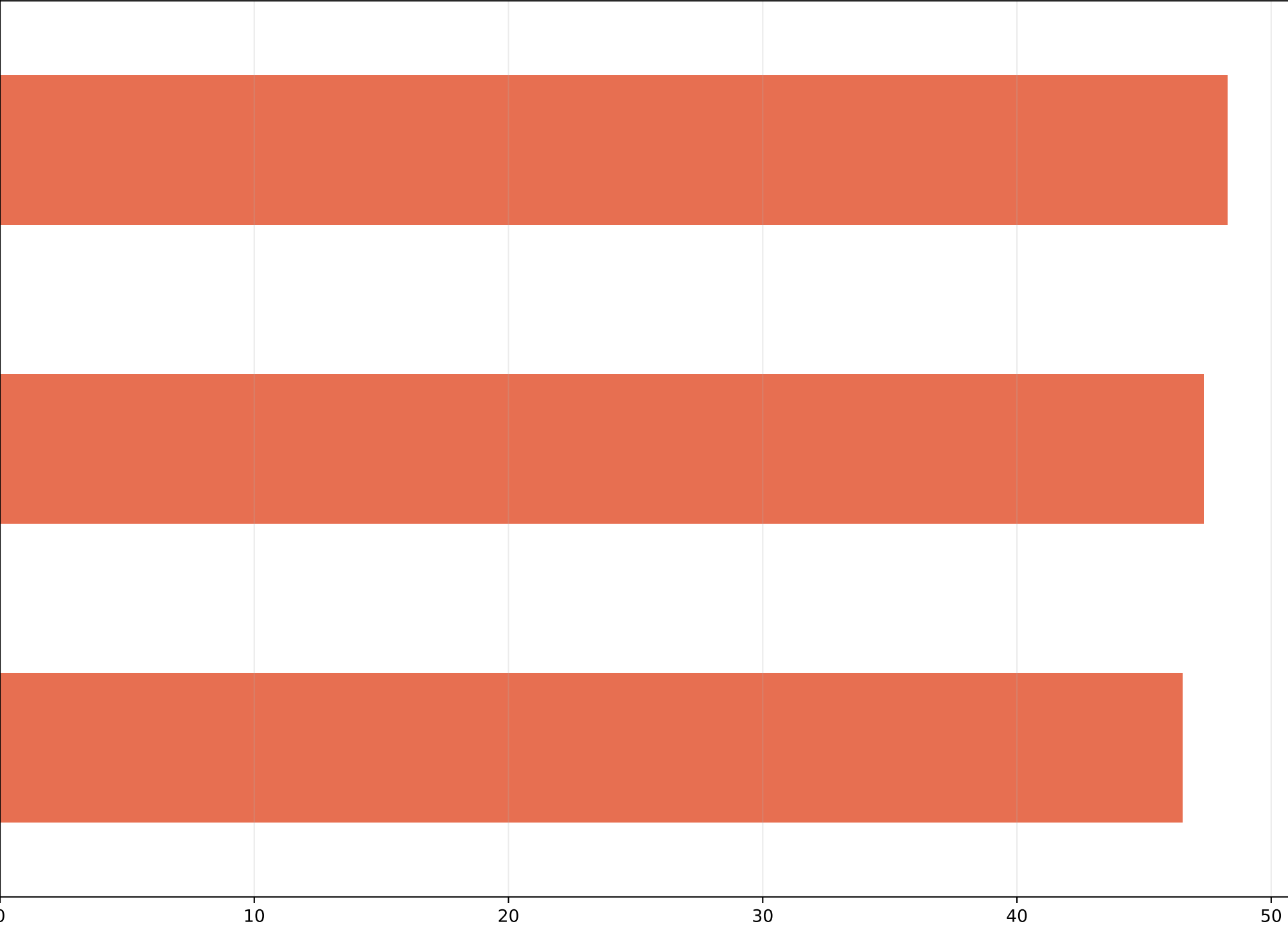
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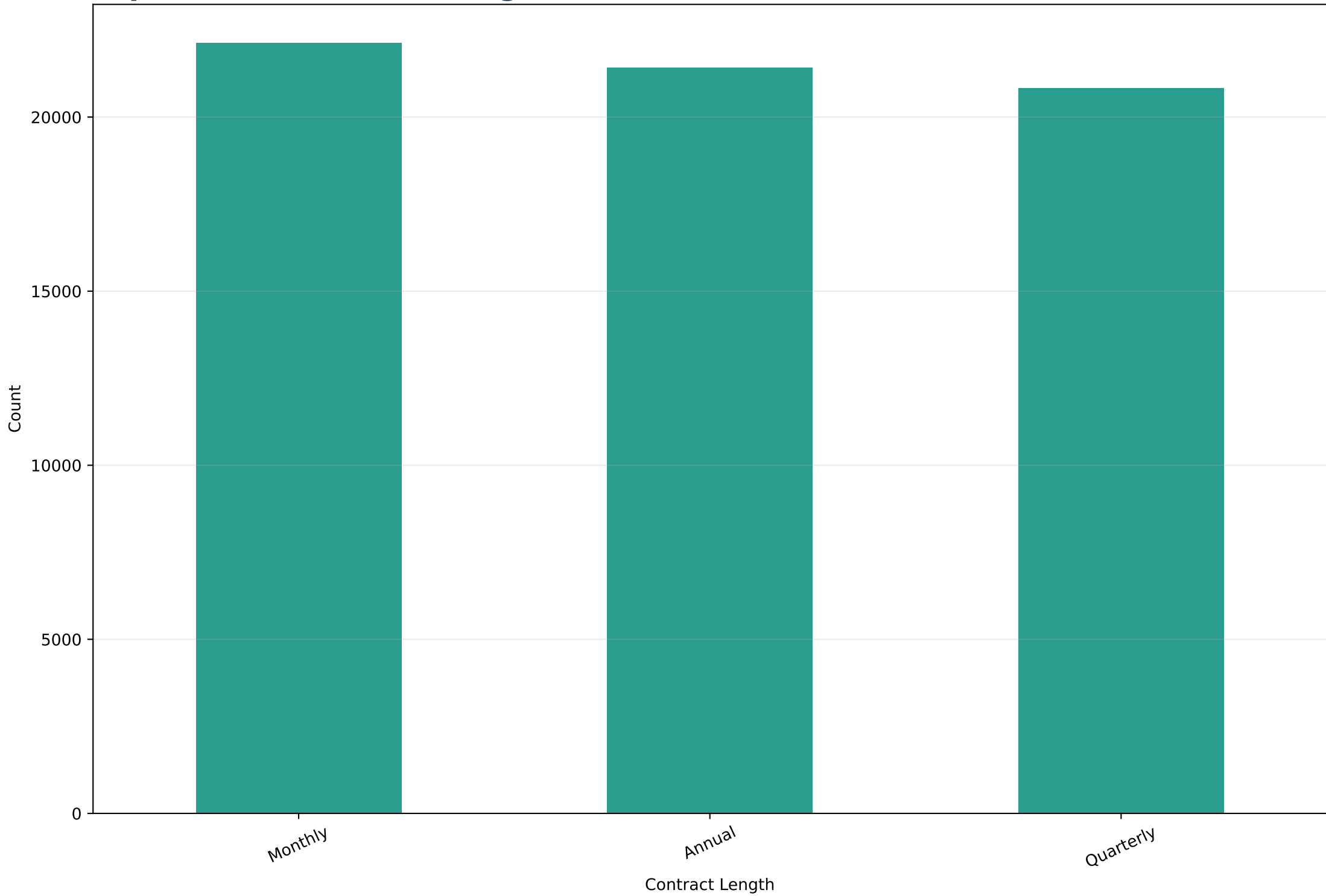
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Churn rate (%)



Top Values: Contract Length



Churn Rate Detail: Contract Length

Contract Length	Customers	Churned	Churn rate %
Annual	21410	9895	46.22
Monthly	22130	11421	51.61
Quarterly	20834	9177	44.05

Churn Rate by Contract Length

Contract Length

Monthly

Annual

Quarterly

0

10

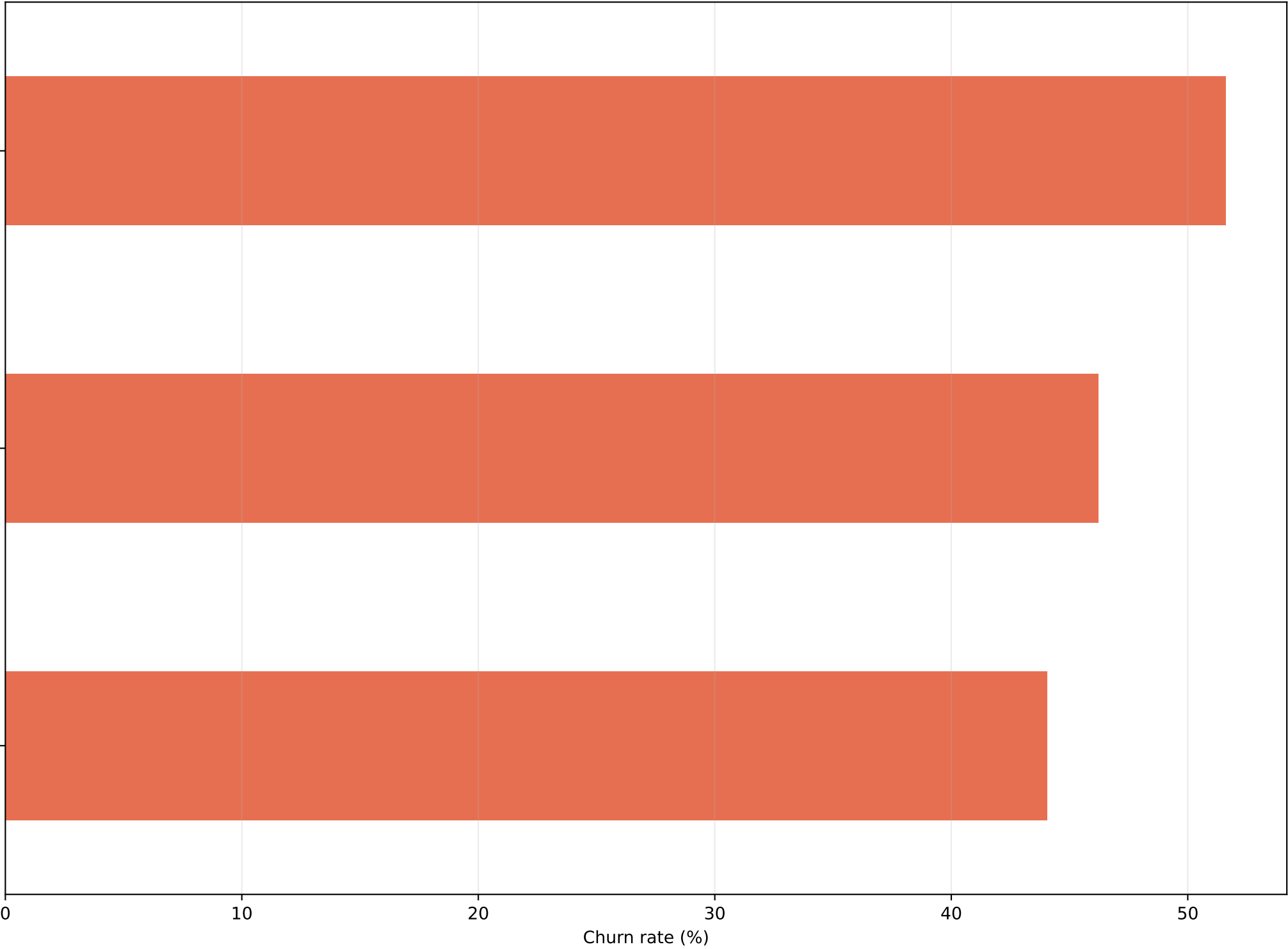
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30

40

50

Churn rate (%)



Target Balance

Churn value	Customers	Percentage
0	33881	52.63
1	30493	47.37

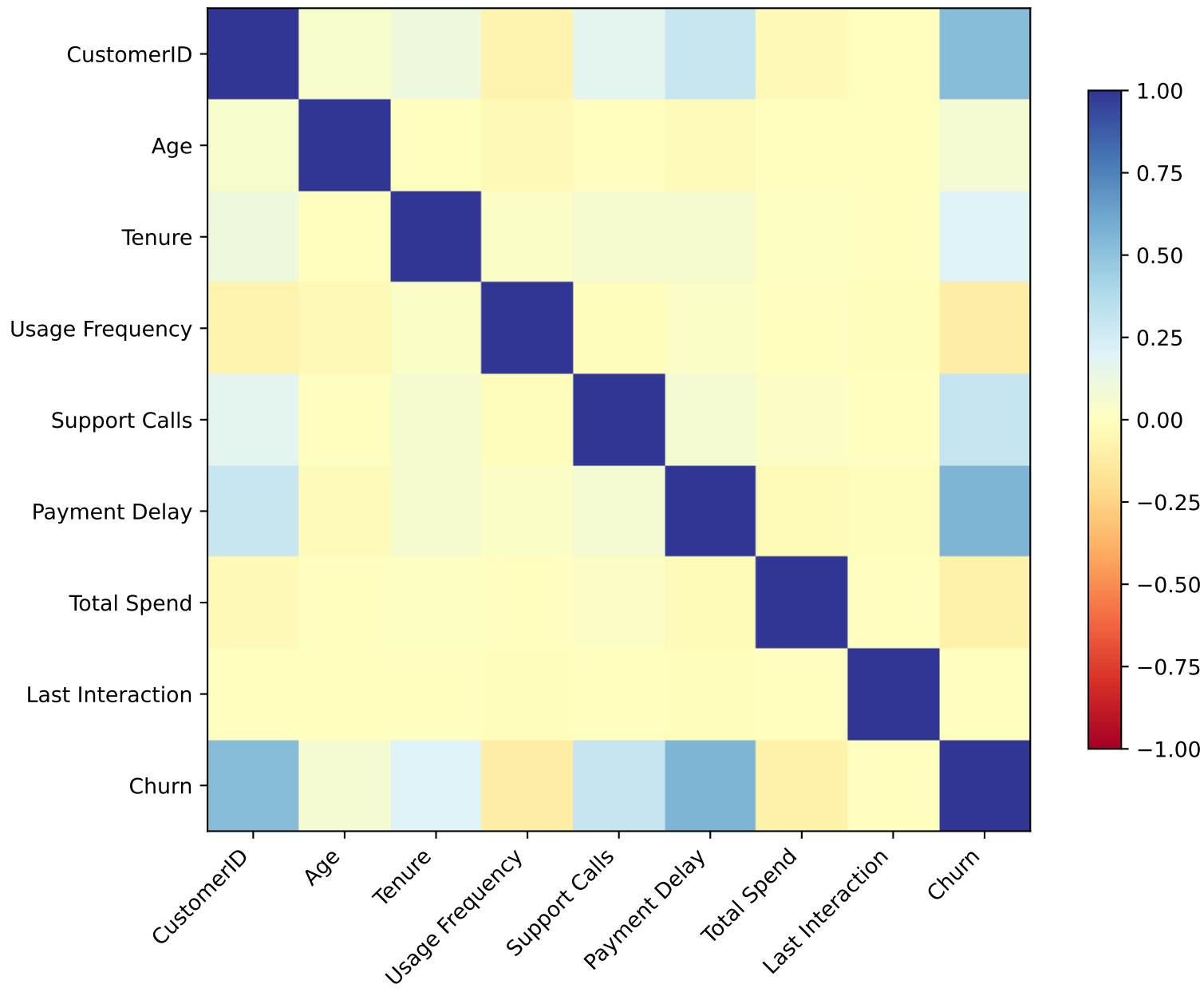
Customer ID Integrity

Check	Result
Unique CustomerID values	64374
Repeated CustomerID rows	0
Minimum CustomerID	1
Maximum CustomerID	64374
IDs form 1..row count	True

Potential Numeric Outliers (IQR Rule)

Column	Lower bound	Upper bound	IQR outlier count
CustomerID	-32185.5	96560.5	0
Age	-6.0	90.0	0
Tenure	-25.5	90.5	0
Usage Frequency	-17.0	47.0	0
Support Calls	-4.5	15.5	0
Payment Delay	-12.5	47.5	0
Total Spend	-369.5	1450.5	0
Last Interaction	-14.5	45.5	0
Churn	-1.5	2.5	0

Numeric Feature Correlations



Features Most Associated with Churn

Feature	Absolute correlation with Churn	Signed correlation
Payment Delay	0.56	0.56
CustomerID	0.53	0.53
Support Calls	0.3	0.3
Tenure	0.2	0.2
Usage Frequency	0.12	-0.12
Total Spend	0.08	-0.08
Age	0.06	0.06
Last Interaction	0.0	-0.0

Exploration Findings

The dataset has 64,374 rows and 12 columns.

Total null values: 0.

Total blank-string values: 0.

Duplicate rows found: 0.

Target column check: Churn is present.

CustomerID integrity: IDs are unique and sequential.

Review the distribution pages for imbalances and the IQR page for values that may require domain validation.

Churn-rate pages compare the target outcome across categorical segments; correlation is association, not causation.

This report is descriptive: it does not remove, impute, or modify any source data.